PROJECT TITLE

Explainable AI-Driven Multimodal Analytics for Sustainable Smart City Decision Intelligence

BACHELOR OF TECHNOLOGY
In
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

The mini-project report entitled "Explainable AI-Driven Multimodal Analytics for Sustainable Smart City Decision Intelligence" is a record of bona fide work carried out by Sharmila Pathi – 2420030042, Vishnu Priya Kodali – 2420030630, Ivaturi Vennela – 2420030185, Lakshman Angara – 2420030067 in partial fulfillment of the requirements for the award of the B.Tech degree at K L (Deemed to be) University. The work presented in this report is original to the best of our knowledge, and all datasets, tools, code, and external sources used have been appropriately acknowledged.

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CERTIFICATE

This is to certify that the mini-project report entitled "Explainable AI-Driven Multimodal Analytics for Sustainable Smart City Decision Intelligence" is a bona fide work carried out by Sharmila Pathi – 2420030042, Vishnu Priya Kodali – 2420030630, Ivaturi Vennela – 2420030185, Lakshman Angara – 2420030067 under the supervision of Dr. K Swapnika during the academic year [2026-2027], in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

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1. ABSTRACT

Rapid urbanization has resulted in several challenges, including increasing air pollution, unpredictable weather conditions, and rising energy consumption. These factors have a significant impact on environmental sustainability and urban planning. Managing these interconnected issues efficiently has become a major challenge for developing sustainable smart cities.

Existing smart city applications mainly focus on monitoring individual factors such as air quality, weather, or energy consumption separately. However, they often lack an integrated approach and provide limited insights into the relationship between these domains. This creates the need for a unified and explainable data analytics system that supports informed decision-making.

The proposed project, "Explainable Data Analytics for Sustainable Smart City Applications," integrates Air Quality, Historical Weather, and Energy Consumption datasets to analyze urban sustainability. The system performs data preprocessing, exploratory data analysis, and predictive modeling using Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor. SHAP (SHapley Additive Explanations) is used to improve the transparency and interpretability of the model predictions.

In the future, the proposed system can be extended by integrating real-time IoT sensor data, supporting multiple cities, and incorporating advanced deep learning techniques to enhance prediction accuracy. It can also be expanded into a comprehensive Smart City Decision Support System for sustainable urban planning and efficient resource management.

Team Name: EcoMinds

Team Logo (optional):



Team Members and Responsibilities:

1. [Sharmila Pathi, 2420030042. – Conceptualization, methodology, data preprocessing, and model development.]

2. [Vishnu Priya Kodali, 2420030630. – Data analysis, visualization, and results evaluation.]

3. [Ivaturi Vennela, 2420030185. – Explainable AI analysis, documentation, and manuscript preparation.]

4. [Lakshman Angara, 2420030067. – Literature review, validation, and manuscript review.]

Dataset/source and project repository

Datasets

1. **Air Quality Monitoring Dataset *(Environmental and Air Pollution Analytics)***

   Source: https://www.kaggle.com/datasets/ziya07/air-quality-dataset

2. **Historical Weather Dataset (Climate and Weather Analytics)**

   Source: https://www.kaggle.com/datasets/zulqarnain11/synthetic-weather-dataset

3. **Energy Consumption Dataset (Smart Energy Analytics)**

   Source: https://www.kaggle.com/datasets/mrsimple07/energy-consumption-prediction

Project Repository

https://github.com/sharmi-pathi/KLH-CSE-2026-27-2420030042-Explainable-Data-Analytics-for-Sustainable-Smart-City-Applications

https://github.com/VishnuPriyaKodali/KLH-CSE-2026-27-2420030630-Explainable-Data-Analytics-for-Sustainable-Smart-City-Applications

https://github.com/vennelaa27/KLH-CSE-2026-27-2420030185-Explainable-Data-Analytics-for-Sustainable-Smart-City-Applications

https://github.com/lakshman067/KLH-CSE-2026-27-2420030067-Explainable-Data-Analytics-for-Sustainable-Smart-City-Applications

2. INTRODUCTION

Rapid urbanization has intensified the challenges associated with environmental pollution, changing weather conditions, and increasing energy consumption. These factors play a significant role in urban sustainability and directly influence the quality of life, resource management, and environmental health of cities. With the growing availability of urban data, data analytics and machine learning provide opportunities to identify patterns, predict future conditions, and support informed decision-making. However, these domains are frequently analyzed independently, limiting the ability to understand their relationships and derive comprehensive insights for sustainable smart-city planning.

The proposed project, **"Explainable Data Analytics for Sustainable Smart City Applications"** develops an integrated analytical framework that combines air-quality, historical weather, and energy-consumption data within a unified workflow. The methodology incorporates data preprocessing, exploratory data analysis, correlation analysis, feature engineering, feature selection, and comparative evaluation of machine-learning models. The framework employs the proposed Out-of-Fold Linear–CatBoost–Gradient Boosting Stacking Ensemble (LCG-Stack) with a Linear Regression meta-learner to perform predictive analysis across the three domains.

Furthermore, Explainable Artificial Intelligence (XAI), particularly SHAP (SHapley Additive exPlanations), is incorporated to interpret model predictions and identify influential features. Cross-domain analytics are used to explore relationships between environmental conditions and energy-related patterns, while an interactive Streamlit dashboard presents analytical results, model performance, predictions, and explainability insights in an accessible format.

The primary objective of this project is to establish a reliable, interpretable, and integrated data analytics framework that supports evidence-based urban planning. By combining predictive modeling, explainability, cross-domain analysis, and interactive visualization, the proposed framework aims to assist urban planners, environmental analysts, and other stakeholders in making informed decisions toward sustainable smart-city development.

3. LITERATURE SURVEY

1. **Interpretable Machine Learning Framework for Air Quality Prediction in Istanbul using SHAP (2026)**

   Develops ML models for predicting major air pollutants and uses SHAP to identify how meteorological and pollutant variables influence predictions.

   https://link.springer.com/article/10.1007/s00477-026-03168-4

2. **Deep Learning-Based AQI Forecasting: A CNN-LSTM Model with Visual insights from SHAP-LIME and PDP (2025)**

   Uses a CNN-LSTM model for AQI forecasting and applies SHAP, LIME, and PDP to explain the factors affecting predictions.

   https://link.springer.com/article/10.1007/s42452-025-07845-x

3. **Scalable AI-Driven Air Quality Forecasting and Classification for Public Health Applications (2025)**

   Combines ML/DL models with meteorological data for air-quality forecasting and uses SHAP and LIME to provide interpretable results.

   https://link.springer.com/article/10.1007/s44292-025-00052-8

4. **Machine Learning Model for Predicting Long-Term Energy Consumption in Buildings (2025)**

   Compares XGBoost, SVR, and LSTM models for predicting long-term building energy consumption and evaluates their forecasting performance.

   https://link.springer.com/article/10.1007/s43926-025-00115-7

5. **Air Quality Forecasting Using Machine Learning: Comparative Analysis and Ensemble Strategies for Enhanced Prediction (2025)**

   Compares multiple machine-learning models, including XGBoost, Random Forest, CatBoost, and LightGBM, for improved air-quality prediction.

   https://link.springer.com/article/10.1007/s11270-025-08122-8

6. **Comprehensive Evaluation of Machine Learning Models for Real-World Air Quality Prediction and Health Risk Assessment by AirQ+ (2025)**

Evaluates different ML models for particulate-matter prediction using pollutant and meteorological variables, with applications to health-risk assessment.

   https://link.springer.com/article/10.1007/s12145-025-01941-7

7. **Smart Cities, Hot Cities: ML-Based Forecasting of Urban Heat Patterns (2025)**

   Applies machine learning to forecast urban temperature patterns and uses SHAP to explain the influence of climatic and temporal factors.

   https://link.springer.com/article/10.1007/s43621-025-02059-y

8. **Explainable Energy Consumption and Speed Prediction in Sustainable Cities Using Deep Learning (2025)**

   Uses deep-learning techniques to predict energy consumption and incorporates Explainable AI to interpret the model's predictions.

   https://link.springer.com/article/10.1007/s00521-024-10850-7

9. **A Machine Learning-Powered Energy Consumption Prediction System with API (2025)**

   Develops an ML-based energy prediction system with an API, demonstrating how predictive models can be integrated into practical applications.

   https://link.springer.com/article/10.1186/s43067-025-00242-9

10. **State-of-the-Art and Emerging Trends in Deep Learning for Air Quality Forecasting and Health Impact Modeling (2025)**

    Reviews recent deep-learning approaches for air-quality forecasting and examines emerging technologies for environmental and health-impact monitoring.

    https://ieeexplore.ieee.org/abstract/document/11410867

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

Rapid urbanization has created interconnected challenges such as increasing air pollution, changing weather conditions, and rising energy consumption. Existing analytical solutions often study these domains independently, making it difficult to understand their relationships and provide comprehensive insights for sustainable urban planning. Furthermore, machine learning models may provide predictions without clearly explaining the factors influencing their results. Therefore, there is a need for an integrated and explainable data analytics framework that can analyze environmental, weather, and energy-related data, generate reliable predictions, and provide understandable insights to support sustainable smart city decision-making.

Problem Description:

Rapid urbanization increases the challenges of air pollution, changing weather conditions, and energy consumption, making sustainable urban planning increasingly important. Existing approaches often analyze these domains separately and provide limited explanations for prediction results. Urban planners, environmental analysts, and city administrators need an integrated, explainable analytics framework that delivers reliable predictions and actionable insights for informed smart-city decision-making.

Objectives:

- **To develop** an interactive analytics platform for analyzing air quality, weather, and energy consumption data to support sustainable smart-city decision-making.
- **To integrate** air quality, weather, and energy consumption datasets to identify cross-domain relationships and meaningful patterns.
- **To enhance** predictive analytics by incorporating Explainable AI (SHAP) to identify and interpret the key factors influencing environmental and energy predictions.
- **To evaluate** and compare machine-learning models for environmental and energy prediction using appropriate performance metrics and identify the most effective models.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- A computer or laptop with a modern multi-core processor.
- Minimum 8 GB RAM recommended for data processing, model training, and dashboard execution.
- Sufficient storage space for datasets, Jupyter notebooks, trained models, visualizations, and SHAP results.
- Internet connectivity for downloading datasets, installing libraries, and accessing online resources.
- GPU is optional, as the project primarily uses classical machine learning algorithms.

Software Requirements:

- Programming Language: Python 3.x.
- Development Environment: Jupyter Notebook and PyCharm.
- Data Processing: Pandas, NumPy.
- Machine Learning: Scikit-learn, CatBoost.
- Data Visualization: Matplotlib, Seaborn.
- Explainable AI: SHAP.
- Dashboard Development: Streamlit.
- Dataset Format: CSV files containing air quality, weather, and energy consumption data.
- Version Control: Git and GitHub for source code management and project documentation.

6. IMPLEMENTATION

A. Project Workflow and Modules

The proposed project, Explainable Data Analytics for Sustainable Smart City Applications, integrates air quality, weather, and energy consumption data to perform preprocessing, exploratory analysis, feature engineering, predictive modeling, and explainability.

1. Data and Task: The project uses three datasets representing air quality, weather conditions, and energy consumption. The tasks include predicting Pollution_Score for air quality, Temperature (C) for weather, and Energy Consumption for energy. The datasets are stored in CSV format. The project uses regression models, and the dataset sources and applicable usage conditions should be documented in the repository.

2. Data Preprocessing and Feature Engineering: Missing values, duplicate records, and inconsistent data are handled during preprocessing. Feature engineering and feature selection are performed to improve data representation and model efficiency. Target-related leakage features are excluded to ensure fair evaluation. Since this is a structured-data analytics project rather than an NLP project, text tokenization, stop-word removal, stemming, lemmatization, and TF-IDF are not applicable.

3. Model Training and Evaluation: Linear Regression, Decision Tree, Random Forest, and the proposed LCG-Stack ensemble are used for model evaluation. The ensemble combines Linear Regression, CatBoost, and Gradient Boosting as base learners, with a Linear Regression meta-learner trained using out-of-fold predictions. Models are evaluated using MAE, MSE, RMSE, R^2, and other regression metrics. Preprocessing and model selection follow a leakage-aware workflow, and the final model is assessed using held-out test data.

Project Modules

- Data Loading and Exploratory Analysis: DAV_Analysis.ipynb loads the datasets and explores distributions, correlations, and patterns.
- Data Cleaning and Preprocessing: Handles missing values, duplicates, and dataset-specific preprocessing requirements.
- Feature Engineering and Selection: Creates relevant features and selects suitable predictors for each domain.
- Feature Extraction and Model Training: Implements regression models and the LCG-Stack ensemble.

- Evaluation and Explainability: Evaluates model performance, generates visualizations, and uses SHAP to explain feature contributions.
- Cross-Domain Analytics: Combines relevant information across domains to explore relationships between urban environmental factors.
- Dashboard and Smart-City Insights: dashboard.py provides an interactive interface for analysis, predictions, explainability, and decision support.

B. Machine-Learning Implementation

The project compares conventional regression models with the proposed stacking ensemble.

Model	**Purpose**
Linear Regression	Baseline regression model
Decision Tree Regressor	Captures nonlinear patterns using tree-based rules
Random Forest Regressor	Combines multiple decision trees to improve predictive robustness
LCG-Stack	Combines Linear Regression, CatBoost, and Gradient Boosting using a Linear Regression meta-learner

The LCG-Stack model uses out-of-fold predictions to train its meta-learner, helping reduce leakage during ensemble training. SHAP is applied to explain how individual features contribute to model predictions. The implementation, preprocessing workflow, model evaluation, and explainability results are documented in the Jupyter notebook and supporting project folders.

C. Demonstration / Dashboard

The project includes an interactive Streamlit dashboard, implemented in dashboard.py. It brings together the three analytical domains and provides a user-friendly interface for exploring datasets, examining visualizations, comparing model performance, generating predictions, and understanding model outputs through SHAP explanations. Cross-domain analysis and smart-city decision-support insights further help users interpret the results. The dashboard serves as a demonstration interface for the integrated analytics framework.

7. RESULTS AND DISCUSSION

The proposed Explainable Data Analytics for Sustainable Smart City Applications framework was implemented and evaluated across three domains: air quality, weather, and energy consumption. Linear Regression, Decision Tree, and Random Forest models were compared with the proposed LCG-Stack ensemble, which combines Linear Regression, CatBoost, and Gradient Boosting using a Linear Regression meta-learner trained with five-fold out-of-fold predictions.

Model performance was evaluated using MAE, MSE, RMSE, R^2, Median Absolute Error, Explained Variance, Maximum Error, NRMSE, Adjusted R^2, and MAPE where applicable. The results showed that model performance varied across domains. The proposed ensemble improved upon Linear Regression for weather prediction but did not outperform it for air quality and energy consumption across the reported RMSE and R^2 metrics. This indicates that model effectiveness depends on dataset characteristics and that ensemble methods do not always outperform simpler models.

SHAP-based explainability was used to identify influential features and understand their contributions to model predictions through global and individual-level explanations. Cross-domain analysis was also performed to explore relationships between environmental conditions, weather patterns, and energy consumption.

The completed Streamlit dashboard integrates data exploration, visualizations, model evaluation, predictions, and explainability results. Overall, the framework demonstrates the value of combining predictive modeling, explainable AI, and cross-domain analytics to generate interpretable insights and support data-driven sustainable smart-city decision-making.

8. CONCLUSION AND FUTURE WORK

The study developed and evaluated an explainable data analytics framework for sustainable smart-city applications by integrating heterogeneous air quality, weather, and energy-consumption data. The framework incorporated data preprocessing, exploratory analysis, feature engineering, feature selection, comparative machine-learning modeling, hybrid predictive modeling, SHAP-based explainability, cross-domain analysis, and scenario-based insights.

Linear Regression, Decision Tree, and Random Forest were evaluated as baseline models and compared with the proposed hybrid model consisting of Linear Regression, CatBoost, and Gradient Boosting combined through a 5-fold out-of-fold stacking procedure with a Linear Regression meta-learner. The experimental results showed that model performance varied across the three domains. The proposed hybrid model achieved competitive performance but did not consistently outperform the baseline models, demonstrating that model effectiveness depends on the characteristics of the underlying dataset and domain.

SHAP analysis provided global and instance-level explanations of the proposed hybrid model, helping identify the features contributing to predictions across the three domains. Cross-domain and scenario-based analyses further provided insights into relationships among environmental and energy-related variables without making causal claims. These analytical components were integrated into an interactive Streamlit dashboard that combines predictions, explanations, cross-domain evidence, and decision-support recommendations.

Future work can extend the framework by incorporating larger, more diverse, and geographically distributed datasets to improve generalizability. Real-world sensor and IoT data with reliable timestamps could enable more comprehensive temporal and spatial integration across smart-city domains. Future studies may also investigate additional advanced ensemble and hybrid modeling techniques, systematic hyperparameter optimization, and alternative explainability methods to further evaluate model robustness and interpretability.

The cross-domain component can be extended to include additional urban domains such as traffic, water management, waste management, and renewable-energy systems. Future research could also investigate causal analysis and temporal forecasting where suitable data and methodological validation are available. Finally, the dashboard could be extended with continuously updated data sources and more advanced decision-support capabilities while maintaining transparency, interpretability, and human oversight.

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